

Memory MCP on AWS EC2 — Setup Q&A

Review guide covering common issues while deploying **Memory MCP** (`memory_mcp.server:app`) on **Amazon Linux EC2**, with **nginx**, **Cloudflare** (`mcp.livemigrate.ai`), **Let's Encrypt**, **Google OAuth**, and **Upstash Redis**.

Architecture & URLs

Q: What is the public MCP endpoint URL?

A: With default env:

- `MCP_MOUNT_PREFIX=/api`
- `MCP_PATH=/mcp`

Clients use:

```
https://mcp.livemigrate.ai/api/mcp
```

Health check:

```
https://mcp.livemigrate.ai/api/health
```

(`GET` should return `OK` .)

Q: Why does `curl http://127.0.0.1:8007/health` return nothing or 404?

A: Health is registered on the FastMCP app **under the mount prefix**, not at the Starlette root. With `MCP_MOUNT_PREFIX=/api` , use:

```
curl -s http://127.0.0.1:8007/api/health
```

Plain `/health` returns **404 Not Found** (not an empty body). Exit code `000` from curl usually means **nothing is listening** on 8007 (service down).

Q: What should `BASE_URL` be in `/etc/mcp-servers/env` ?

A: The URL users and OAuth see in the browser — **scheme + host**, no path:

```
BASE_URL=https://mcp.livemigrate.ai
```

Do **not** put `/api` in `BASE_URL` ; the server combines `BASE_URL` + `MCP_MOUNT_PREFIX` for auth base URL internally.

Q: Why must nginx proxy `/` and not only `/api/` ?

A: The Starlette app mounts OAuth **.well-known** routes at the **root**. MCP tools live under `/api/*` . A config that only proxies `/api/` breaks OAuth discovery and login. The repo nginx config proxies **location** `/` to Uvicorn.

Environment file on EC2

Q: Why does `source /etc/mcp-servers/env` fail with “Permission denied”?

A: The file is `chmod 600` and owned by `root`. SSM users (`ssm-user`) cannot read it.

You do not need to source it for production: `systemd` reads `EnvironmentFile=` as root before dropping to `User=ec2-user` .

For a one-off manual test as root:

```
sudo bash -c 'set -a && . /etc/mcp-servers/env && set +a && cd /opt/mcp-servers &&
runuser -u ec2-user -- env PATH="/opt/mcp-servers/.venv/bin:$PATH"
PYTHONPATH=/opt/mcp-servers/src ...'
```

Q: Should I use `.env` in the repo on EC2?

A: No. Use `/etc/mcp-servers/env` only (see `deploy/aws-ec2/env.example`). Never commit secrets.

Q: What env vars are required for the server to start?

A: At minimum:

- `OAUTH_CLIENT_ID` , `OAUTH_CLIENT_SECRET`
- `JWT_SIGNING_KEY` , `STORAGE_ENCRYPTION_KEY`
- `REDIS_URL` (Upstash `rediss://` TCP URL, not REST)
- `BASE_URL`
- `MCP_MOUNT_PREFIX=/api` and usually `MCP_PATH=/mcp`

Missing vars cause **import-time crash** with `RuntimeError: Missing required environment variable ...` .

systemd & Uvicorn

Q: What does `status=217/USER` mean?

A: The `User=` in the unit file does not exist on the OS. Deploy units default to `ec2-user` on Amazon Linux. `www-data` exists on Ubuntu, not AL2023.

Fix: set `User=ec2-user` / `Group=ec2-user` and `chown -R ec2-user:ec2-user /opt/mcp-servers` .

Q: What does `status=203/EXEC` or “Failed to execute ... uvicorn: Permission denied” mean?

A: `systemd` could not run `ExecStart` . Common causes:

1. **Wrong unit** still calling `/opt/mcp-servers/.venv/bin/uvicorn` instead of `python -m uvicorn`
2. **Venv built as `ssm-user`** but service runs as `ec2-user` — `.venv/bin/python` symlinks to `/home/ssm-user/.local/share/uv/...` and `ec2-user` cannot execute it

Fix venv ownership:

```
sudo -u ec2-user bash -lc 'cd /opt/mcp-servers && rm -rf .venv && uv sync --frozen --no-dev'
```

Use in unit:

```
ExecStart=/opt/mcp-servers/.venv/bin/python -m uvicorn memory_mcp.server:app --host 127.0.0.1 --port 8007
Environment=PYTHONPATH=/opt/mcp-servers/src
```

Rule: Run `uv sync` as the **same user** as `User=` in systemd (`ec2-user`), not from SSM as `ssm-user` .

Q: Direct vs nginx systemd unit?

A:

Mode	Unit	Uvicorn bind
ALB / lab on :8007	memory-mcp.service	0.0.0.0:8007
nginx on instance	memory-mcp-nginx.service	127.0.0.1:8007

With nginx, do **not** expose 8007 on the security group.

Git on EC2

Q: `git pull` — “dubious ownership” and “cannot open .git/FETCH_HEAD: Permission denied”?

A: Repo owned by `ec2-user` , you are `ssm-user` .

```
git config --global --add safe.directory /opt/mcp-servers # fixes dubious ownership warning only
sudo -u ec2-user git -C /opt/mcp-servers pull # actual pull
```

nginx & TLS

Q: `nginx -t` fails — cannot load certificate ... fullchain.pem?

A: `memory-mcp.conf` references Let’s Encrypt paths **before** certbot has created them. Order matters:

1. `memory-mcp.bootstrap.conf` (HTTP only on :80)
2. `certbot certonly --webroot`
3. `memory-mcp.conf` (HTTPS on :443)
4. `sudo nginx -t && sudo systemctl reload nginx`

Always run `sudo nginx -t` (not as unprivileged user).

Q: Cloudflare Origin cert vs Let’s Encrypt on nginx — which is recommended?

A: Let’s Encrypt on nginx + Cloudflare **Full (strict)**:

- Publicly trusted cert on the origin
- Works if you later change DNS/proxy
- Standard certbot renewal

Cloudflare Origin cert is optional when you cannot open port 80 or want fastest paste-and-go PEM setup.

Q: Cloudflare DNS for `mcp.livemigrate.ai` ?

A: A record `mcp` → EC2 public IP. Proxied (orange cloud) is OK with **Full (strict)** once origin has a valid cert.

Security group: **443** (and **80** for HTTP-01 + redirect). Block **8007** from the internet when using nginx.

Q: What is `memory-mcp.bootstrap.conf` for?

A: Serves HTTP on port **80**, proxies to Uvicorn, exposes `/.well-known/acme-challenge/` for certbot. Use until LE files exist under `/etc/letsencrypt/live/mcp.livemigrate.ai/`.

Redis (Upstash)

Q: What Redis URL format?

A: `rediss://` (TLS) TCP endpoint from Upstash, not the REST URL. Server stores OAuth client state in Redis via `RedisStore` + Fernet encryption wrapper.

Q: Local `memory-client` failed connecting to `localhost:6379` ?

A: Client used `RedisStore` without `REDIS_URL`, defaulting to local Redis. Fix: use Redis only when `REDIS_URL` is set; otherwise FastMCP uses **in-memory** token storage.

```
uv run memory-client
uv run --env-file .env memory-client # persistent tokens via Upstash
```

Optional: `MCP_CLIENT_URL=https://mcp.livemigrate.ai/api/mcp`

OAuth & clients

Q: Google OAuth — what to configure?

A: In Google Cloud Console for the same client ID as EC2 env:

- **Authorized JavaScript origins:** `https://mcp.livemigrate.ai`
- **Redirect URIs:** match FastMCP/Google provider (often under `https://mcp.livemigrate.ai/api/...`). Add exact URI from the first browser error if mismatch.

Restart `memory-mcp` after changing `BASE_URL`.

Q: Claude / remote MCP URL?

A: `https://mcp.livemigrate.ai/api/mcp` with OAuth — not stdio. Stdio bypasses HTTP OAuth.

Q: Verify OAuth metadata?

A:

```
curl -sI https://mcp.livemigrate.ai/.well-known/oauth-authorization-server
curl -sI https://mcp.livemigrate.ai/.well-known/oauth-protected-resource
```

Should not be nginx **404**.

End-to-end checklist

1. `uv sync` as `ec2-user`; `/etc/mcp-servers/env` complete; `memory-mcp` active
 2. `curl -s http://127.0.0.1:8007/api/health` → OK
 3. nginx bootstrap → certbot → full `memory-mcp.conf` → reload
 4. `curl -s https://mcp.livemigrate.ai/api/health` → OK
 5. Cloudflare **Full (strict)**
 6. Google OAuth URIs; connect Claude to `/api/mcp`
 7. `sudo certbot renew --dry-run` ; `systemctl enable memory-mcp nginx`
-

Quick reference commands

```
# Service
sudo systemctl restart memory-mcp
sudo journalctl -u memory-mcp -n 50 --no-pager

# App update
sudo -u ec2-user git -C /opt/mcp-servers pull

# nginx
sudo nginx -t && sudo systemctl reload nginx

# Health
curl -s https://mcp.livemigrate.ai/api/health
```

Generated from deployment practice for the `mcp-starter` Memory MCP project. Repo paths: `deploy/aws-ec2/` , `src/memory_mcp/` .